

L-13 chapter- 5.1

Linear Equations

Linear Eqⁿ: All the unknown variables must have power of 1. [x^2, y^3, x^4 - Not allowed]

Some linear equations

* x, y, z are variables.

* there circles are coefficient

$$\begin{cases} 2x + 3y - z = 5 \\ 4x + 1y + 2z = 6 \\ -2x + 5y + 3z = -4 \end{cases}$$

bias/constant

We can solve it and find x, y, z values using Matrix

Some terms,

$$2x + 3y + z = 1$$

there are coefficient

bias/constant

x, y, z are unknown values.

Converting to matrix

$$\begin{bmatrix} 2 & 3 & -1 \\ 4 & 1 & 2 \\ -2 & 5 & 3 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ -4 \end{bmatrix}$$

A X B

so, $AX = B$

In this way, equations can be solved, if :

i) $\det(A) \neq 0$

ii) $A \cdot A^{-1} = I$ [identical matrix]

Triangular Matrix

upper
Lower

Why need?

- easy to solve
- $\det(A)$ calculation is easy
- use in LU decomposition

format:

3x3 upper triangular Matrix:

Example

$$\begin{bmatrix} 5 & 2 & 1 \\ 0 & 7 & 10 \\ 0 & 0 & 5 \end{bmatrix}_{3 \times 3}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

$\det(A) = a_{11} * a_{22} * a_{33}$

4x4 lower triangular Matrix:

format

Example

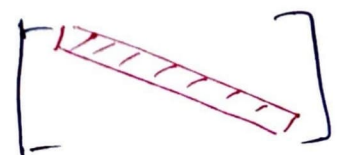
$$\begin{bmatrix} 8 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 \\ -2 & 6 & 5 & 0 \\ 7 & 2 & 3 & -1 \end{bmatrix}_{4 \times 4}$$

$$\begin{bmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$\det(A) = a_{11} * a_{22} * a_{33} * a_{44}$

Remember!

In triangular Matrix, $\det(A)$ = multiply of diagonal values.



An Example using Lower Triangular Matrix

Given

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & 2 \end{bmatrix}; B = \begin{bmatrix} 5 \\ 10 \\ 20 \end{bmatrix}; X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Solⁿ

$$AX = B$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & 2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \\ 20 \end{bmatrix}$$

we top to bottom: 2

$$1 \cdot x = 5$$

$$\therefore x = 5$$

$$2x + 1 \cdot y = 10$$

$$\therefore y = 10 - 2x \xrightarrow{\text{⑤}} 0$$
$$= 0$$

$$3x + 4y + 2z = 20$$

$$\therefore z = \frac{20 - 4y - 3x}{2} \xrightarrow{\text{⑤}} 0$$
$$= 5/2$$

$$\text{so, } x = 5$$

$$y = 0$$

$$z = 5/2$$

Let's see an Example

Solve using upper triangular Matrix:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{bmatrix} ; X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} ; B = \begin{bmatrix} 2 \\ 5 \\ 10 \end{bmatrix}$$

Solⁿ

$$A \cdot X = B$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 10 \end{bmatrix}$$

bottom to top: ↑

$$2z = 10$$

$$\therefore z = 5$$

$$1 \cdot y + 3z = 5 \quad \rightarrow \textcircled{5}$$

$$\therefore y = 5 - 3z \\ = -10$$

$$1 \cdot x + 2y + z = 2$$

$$\therefore x = 2 - 2y + z \quad \rightarrow \textcircled{10} \rightarrow \textcircled{5}$$

$$= 17$$

$$\therefore \begin{bmatrix} x = 17 \\ y = -10 \\ z = 5 \end{bmatrix} \text{ Ans.}$$

Chapter - 5.2

Gaussian Elimination Method

$$AX = B$$

$$\Rightarrow \begin{bmatrix} 5 & \frac{1}{2} & \frac{3}{7} \\ 7 & -2 & -9 \\ -5 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ -8 \end{bmatrix}$$

Calculating & solving this is tough.

So, Gaussian Elimination $\rightarrow$ Makes the A matrix simpler.

$\rightarrow$ Converts A matrix to Triangular Matrix
[upper]

Example

$$2x + y - z = 8$$

$$-3x - y + 2z = -11$$

$$-2x + y + 2z = -3$$

matrix form

$$\begin{bmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ -11 \\ -3 \end{bmatrix}$$

augmented matrix

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ -3 & -1 & 2 & -11 \\ -2 & 1 & 2 & -3 \end{array} \right]$$

augmented matrix
 $[A \text{ matrix} | B]$

now

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ -3 & -1 & 2 & -11 \\ -2 & 1 & 2 & -3 \end{array} \right] \begin{array}{l} \text{--- } R_1 \\ \text{--- } R_2 \\ \text{--- } R_3 \end{array}$$

step 1: Make a_{21} & $a_{31} = 0$ using R_1 .

to do $a_{21} = 0$: $R_2 = R_2 + \left(\frac{3}{2}\right) R_1$

↓ we get

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & 1/2 & 1/2 & 1 \\ -2 & 1 & 2 & -3 \end{array} \right]$$

to do $a_{31} = 0$: $R_3 = R_3 + R_1$

↓

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & 1/2 & 1/2 & 1 \\ 0 & 2 & 1 & 5 \end{array} \right]$$

step 2: Make $a_{32} = 0$ using R_2 .

to do $a_{32} = 0$: $R_3 = R_3 - 4R_2$

↓

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & 1/2 & 1/2 & 1 \\ 0 & 0 & -1 & 1 \end{array} \right]$$

we got
upper
triangular

Now

$$\begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 1 \\ 1 \end{bmatrix}$$

$$z = -1$$

$$0 \cdot x + \frac{1}{2} y + \frac{1}{2} z = 1 \quad \text{---} \textcircled{-1}$$

$$\therefore y = 3$$

$$2 \cdot x + 1 \cdot y - 1 \cdot z = 8$$

$$\therefore x = 2$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$$

